Modeling Dynamical Systems

→ Expression of system's behavior by some mathematical relations obtained by setting reasonable assumptions.

Differential Eq
Integral Eq
Integro diff Eq

Differential Equations
- partial: (continuous system) number of independent variables is more than one
  $a_1 \frac{\partial^2 c}{\partial x^2} + a_2 \frac{\partial^2 c}{\partial y^2} + a_3 \frac{\partial^2 c}{\partial x \partial y} + a_4 \frac{\partial^2 c}{\partial t^2} = r(t)$ (Linear PDE, order=2, degree=1)
- ordinary: (Lumped/discretized) $a_1 \frac{d^2 c}{dt^2} + a_2 \frac{dc}{dt} + a_3 c = r(t)$ (Linear ODE, order=2)

(ODE, 2nd order, 3rd degree) ⟷ $a_1 \frac{d^2 c}{dt^2} + a_2 \frac{dc}{dt} + a_3 c^3 + a_4 c^2 = r(t)$ → nonlinear α

(algebraic)
* Lumped, discretizing

System Dynamics:
Response do to
Initial conditions.

Dynamical system
↳ A system changing
to the time

Systems
- static α
- non-static
  - Dynamic (deterministic) ✓
  - stochastic α

We have to talk about linear ODE's that change respect to time

We use laplace to change differential equation to algebric equation. ($s^2 + s + a = r$ algebric)

Laplace Transform

$$\mathcal{L}[f(t)] = \int_0^{\infty} e^{-st} f(t)\, dt = F(s)$$

$\mathcal{L}$: Laplace operator

Differentiation theorem in laplace transform

$$\mathcal{L}\left(\frac{d^n f(t)}{dt^n}\right) = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \cdots - f^{(n-1)}(0)$$

$$a_0 y^{(n)} + a_1 y^{(n-1)} + a_2 y^{(n-2)} + \cdots + a_n y = b_0 x^{(m)} + b_1 x^{(m-1)} + \cdots + b_m x \qquad (n > m)$$

(y: output, x: input)

(the degree is n) ~~order~~

Most genal ODE for a system.

Assuming that initial conditions are zero (to ignore transient response)

$$a_1 \ddot{x} + a_2 \dot{x} + a_3 x = r(t)$$

(x: output, $r(t)$: input)

$$x(t=0) = x_0$$

$$\dot{x}(t=0) = \dot{x}_0$$

Initial condition: $n-1$ 'ekadar $y^{n-1}(0), y^{(n-2)}(0), \ldots, y(0)$

n tane bulunmakta

Laplace transform

$$(a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_n) Y(s) = (b_0 s^m + b_1 s^{m-1} + b_2 s^{m-2} + \cdots + b_m) X(s)$$

$$\text{Transfer function} = \frac{\text{output}}{\text{Input}} = \frac{Y(s)}{X(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_n} = G(s)$$

this is independent of input (intrensic definition of system)

* The max order of derivave define the output of the system

$$G(s) = \frac{Y(s)}{X(s)} \rightarrow Y(s) = X(s) G(s)$$

response, input

$$\mathcal{L}^{-1} \rightarrow y(t) = \int_0^t x(z)\, g(t-z)\, dz = \int_0^t x(t-z)\, g(z)\, dz$$

Impulse function in frequency domain

$X(s) = 1 \longrightarrow Y(s) = G(s)$

impulse response = T.F. (Transfer function)

$$\mathcal{L}^{-1}[G(s)] = g(t)$$

$$\frac{d^3c}{dt^3} + 3\frac{d^2c}{dt^2} + 7\frac{dc}{dt} + 5c = \frac{d^2r}{dt^2} + \frac{4dr}{dt} + 3r$$

zero initial condition → $\mathcal{L}$

$$\Rightarrow s^3C(s) + 3s^2C(s) + 7sC(s) + 5C(s) = s^2R(s) + 4sR(s) + 3R(s)$$

$$\hookrightarrow \frac{C(s)}{R(s)} = \frac{s^2+4s+3}{s^3+3s^2+7s+5} = G(s) \quad \text{T.F.}$$

→ Dynamical system representation.

$$\Rightarrow C(s) = R(s)\,G(s)$$

Find the D.E for the transfer function

$$G(s) = \frac{2s+1}{s^2+6s+2} = \frac{C(s)}{R(s)} \begin{matrix} \text{output} \\ \text{input} \end{matrix}$$

$$(2s+1)R(s) = C(s)(s^2+6s+2)$$

$\mathcal{L}^{-1}$

$$2\dot{r}(t) + r(t) = \frac{d^2c(t)}{dt^2} + 6\frac{dc(t)}{dt} + 2c(t)$$

Linearity theorem:

$$\mathcal{L}[k\,f(t)] = K\,F(s)$$

$$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s) \quad \text{Superposition.}$$

Freq shift $\quad \mathcal{L}[e^{-at}f(t)] = F(s+a)$

Time shift $\quad \mathcal{L}[f(t-T)] = e^{-sT}F(s)$

Scaling:

$$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$$

Differention:

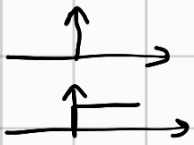
$$\mathcal{L}\left(\frac{d^n f(t)}{dt^n}\right) = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$$

★ **Integration:** $\mathcal{L}\left[\int f(t)\,dt\right] = \frac{F(s)}{s}$

★ **Final value:** $f(\infty) = \lim_{s\to 0} sF(s)$

Initial value $f(0) = \lim_{s\to\infty} sF(s)$

Table to know

$f(t)$	$F(s)$
 impulse $\delta(t)$	1
step $u(t)$	$1/s$
Ramp t	$1/s^2$
$\frac{t^{n-1}}{(n-1)!}$	$1/s^n$
e^{-at}	$\frac{1}{s+a}$
$\frac{e^{-at}t^{n-1}}{(n-1)!}$	$\frac{1}{(s+a)^n}$
$\sin(wt)$	$\frac{w}{s^2+w^2}$
$\cos(wt)$	$\frac{s}{s^2+w^2}$
$\sinh wt$	$\frac{w}{s^2-w^2}$
$\cosh wt$	$\frac{s}{s^2-w^2}$

Partial fraction expansion (PFE)

T.F. $G(s) = \frac{B(s)}{A(s)}$ → numerator → roots of numerator polynomial = zero

→ denemenator → " " denominator " pols

case 1: Poles are distinct and with multiplicity of unity!

tekrarlama → the poles are unique different, not repeat

$$G(s) = \frac{B(s)}{A(s)} = \frac{b_0 + b_1 s + b_2 s^2 + \cdots + b_m s^m}{a_0 + a_1 s + a_2 s^2 + \cdots + a_n s^n}$$

poles are $\lambda_1, \lambda_2, \lambda_3 \cdots \lambda_n$

zeros m

$$= \frac{r_1}{s-\lambda_1} + \frac{r_2}{s-\lambda_2} + \cdots + \frac{r_n}{s-\lambda_n}$$

r_n = residues of the fraction

How to obtain the residues.

$n > m$: strictly proper (for all physical systems) ✓

$n = m$: proper

$n < m$: not proper

Example:

$$\frac{s^2-2}{s^3+3s^2+2s} = \frac{s^2-2}{s(s^2+3s+2)} = \frac{r_1}{s} + \frac{r_2}{s+1} + \frac{r_3}{s+2}$$

Roots are $\begin{cases} s_1 = 0 \\ s_2 = -1 \\ s_3 = -2 \end{cases}$

$$= \frac{r_1(s+1)(s+2) + r_2(s)(s+2) + r_3(s)(s+1)}{s(s+1)(s+2)}$$

$$r_1(s^2+2s+s+2) + r_2(s^2+2s) + r_3(s^2+s) = s^2-2$$

$$\begin{cases} s^0: & 2r_1 = -2 \\ s^1: & 3r_1 + 2r_2 + r_3 = 0 \\ s^2: & r_1 + r_2 + r_3 = 1 \end{cases}$$

★ Second method of residue calculation: $r_k = (s-\lambda_k) F(s) \Big|_{s=\lambda_k}$

$$r_1 = s\,F(s)\Big|_{s=0}$$

$$= \frac{s^2-2}{s^2+3s+2}\Big|_{s=0} = \frac{-2}{2} = -1 \checkmark$$

$$G(s) = \frac{B(s)}{A(s)} \rightarrow \text{roots are } \lambda_k$$

$$r_k = \frac{(s-\lambda_k)\,B(s)}{A(s)}\Big|_{s=\lambda_k} \quad * \text{L'Hopital}$$

$$\lim_{s\to\lambda_k} \frac{\text{derivative of num.}}{\text{derivative of den.}} = \lim_{s\to\lambda_k} \frac{B(s)+(s-\lambda_k)\,B'(s)}{A'(s)}$$

more efficient

$$r_k = \frac{B(s)}{A'(s)}\Big|_{s=\lambda_k}$$

Example

$$v^{(3)} - v = 0 \qquad v(0) = 1,\ \dot{v}(0) = 0,\ \ddot{v}(0) = 0$$

$$\frac{d^3 v}{dt^3} = v^{(3)}$$

$$\mathcal{L} = s^3 V(s) - s^2 v(0) - s\dot{v}(0) - \ddot{v}(0) - V(s) = 0$$

$$\rightarrow V(s)(s^3-1) = s^2 \Rightarrow V(s) = \frac{s^2}{s^3-1}$$

$$\begin{array}{l|l} s^3-1 & s-1 \\ \hline -s^3+s^2 & s^2+s+1 \\ \hline s^2-1 & \\ -s^2+s & \\ \hline s-1 & \end{array}$$

$$s^3-1 = (s-1)(s^2+s+1)$$

$s_1 = 1$ $\qquad$ $s_{2,3} = \frac{-1 \pm j\sqrt{3}}{2}$

three poles of multiplicity one

$$V(s) = \frac{s^2}{s^3-1} = \frac{r_1}{s-1} + \frac{r_2}{s+\frac{1}{2}+\frac{\sqrt{3}}{2}j} + \frac{r_3}{s+\frac{1}{2}-\frac{\sqrt{3}}{2}j}$$

$$r_k = \frac{B(s)}{A'(s)}\Big|_{s=\lambda_k}$$

$B(s) = s^2$

$A(s) = s^3 - 1 \Rightarrow A'(s) = 3s^2$

$$r_k = \frac{s^2}{3s^2}\Big|_{s=\lambda_k}$$

$$r_1 = \frac{(1)^2}{3(1)^2} = \frac{1}{3}$$

$r_2 = 1/3$

$r_3 = 1/3$

$$V(s) = \frac{s^2}{s^3-1} = \frac{1/3}{s-1} + \frac{1/3}{s+\frac{1}{2}+\frac{\sqrt{3}}{2}j} + \frac{1/3}{s+\frac{1}{2}-\frac{\sqrt{3}}{2}j}$$

$$V(t) = \mathcal{L}^{-1}\{V(s)\} = \frac{1}{3}\left[e^{+t} + e^{\left(-\frac{1}{2}-\frac{\sqrt{3}}{2}j\right)t} + e^{\left(-1/2+\sqrt{3}/2\,j\right)t}\right]$$

case 2) Distinct poles of multiplicity k_i

Transfer function $G(s) = \dfrac{B(s)}{(s-\lambda_1)^{k_1}(s-\lambda_2)^{k_2}\cdots(s-\lambda_l)^{k_n}}$

n degree polynomial in the denominator: we expect n roots (poles)

$$= \frac{r_{1,1}}{s-\lambda_1} + \frac{r_{1,2}}{(s-\lambda_1)^2} + \cdots + \frac{r_{1,k_1}}{(s-\lambda_2)^{k_1}} + \frac{r_{2,2}}{(s-\lambda_2)^2} + \cdots + \frac{r_{2,k_2}}{(s-\lambda_2)^{k_2}}$$

$\rightarrow \dfrac{r_{2,1}}{s-\lambda_2}$

$$+ \cdots + \frac{r_{l,1}}{s-\lambda_l} + \frac{r_{l,2}}{(s-\lambda_l)^2} + \cdots + \frac{r_{l,k_n}}{(s-\lambda_l)^{k_n}}$$

$$r_{i,k_i} = F(s)(s-\lambda_i)^{k_i}\Big|_{s=\lambda_i}$$

$$r_{i,k_i-j} = \frac{1}{j!}\frac{d^j}{ds^j}\left(F(s)(s-\lambda_i)^{k_i}\right)\Big|_{s=\lambda_i}$$

Example:

$$F(s) = \frac{1}{s^2(s+1)}$$

$\lambda_1 = \lambda_2 = 0 \rightarrow$ multiplicity 2

$\lambda_3 = -1 \rightarrow$ multiplicity 1

$$= \frac{r_1}{s} + \frac{r_2}{s^2} + \frac{r_3}{s+1}$$

$$r_2 = s^2 \frac{1}{s^2(s+1)}\Big|_{s=0} = 1$$

$$r_1 = \frac{d}{ds}\left(\frac{1}{s+1}\right)\Big|_{s=0} = -\frac{1}{(s+1)^2}\Big|_{s=0} = -1$$

$$r_3 = \frac{B(s)}{A'(s)}\Big|_{s=-1} = \frac{1}{3s^2+2s}\Big|_{s=-1} = 1$$

↑ From case 1

↑ $(s^3+s^2)'$ nin tuleni

case 3 proper and **non proper** T.F.S

↓ $m \leq n$; ↓ $m > n$

$$G(s) = \frac{B(s)}{A(s)} = \frac{b_0 + b_1 s + \cdots + b_m s^m}{a_0 + a_1 s + \cdots + a_n s^n}$$

A make at least some portion of T.F. strictly proper ($n > m$)

$$\frac{s^2+1}{5s^2+3} = \frac{1}{5}\left(\frac{s^2+1}{s^2+3/5}\right) = \frac{1}{5}\left(\frac{s^2+3/5 - 3/5 + 1}{s^2+3/5}\right)$$

$$= \frac{1}{5}\left(1 + \frac{2/5}{s^2+3/5}\right)$$

$$\frac{s^4+s^3-2s^2+1}{s^3+2s^2+s} \quad \text{(nonproper)}$$

$$= \frac{s(s^3+s^2-2s)+1}{s^3+2s^2+s} = \frac{s(s^3+2s^2+s) - s(2s^2+s) + s(s^2-2s)+1}{s^3+2s^2+s}$$

$$= s + \frac{-2s^3 - s^2 + s^3 - 2s^2 + 1}{s^3+2s^2+s} = s - \frac{s^3+3s^2-1}{s^3+2s^2+s}$$

$$= s - \frac{(s^3+2s^2+s) - 2s^2 - s + 3s^2 - 1}{s^3+2s^2+s} = s - 1 + \frac{s^2 - s - 1}{s^3+2s^2+s}$$

↓ $s(s^2+2s+1)$

$g(t) = \delta'(t) + \delta(t) + \cdots$